



B.Tech. School of Computer Engineering

COURSE PLAN: THEORY COURSE

Department :	School of Computer Engineering			
Course Name & code :	Database Systems CSS 2201			Core
Semester & branch :	IV		Common to SCE	
Name of the faculty :	Dr. Roopalakshmi R			
No of contact hours/week:	L	T	P	C
	3	1	0	4

Course Outcomes (COs) to PO, PSO, BL Mapping

At the end of this course, the student should be able to:		No. of Contact Hours	Marks	Program Outcomes (POs)	PSOs	BL (Recommended)
CO1	Articulate the usage of database, relational model and organization of files for query based systems.	13	26	1,2,3,4,5	1,2,4	2,3,4
CO2	Write and optimize structured query languages for efficient data retrieval, manipulation and management.	12	25	1,2,3,4,5	1,2,4	2,3,4,5
CO3	Design databases using E-R models for a real world scenario.	6	12	1,2,3,4,5	1,2,4	2,3,4,5
CO4	Apply the normalization technique to decompose given relation schema into effective schemas.	8	16	1,2,3,4,5	1,2,4	2,3,4
CO5	Interpret the various transaction management, recovery and concurrency control techniques.	9	16	1,2,3,4,5	1,2,4	2,3
CO6	Explore the latest database systems.	SDL	5	1,2,3,4,5,12	1,2,4	2,3,4
Total		48	100			

Lesson Plan

L No	Topics	CO Addressed
0	<i>To address OBE philosophy towards NBA and IET accreditation followed by a discussion on Course Plan, Course Outcomes, CO-PO Mapping and Blooms Taxonomy, with prerequisites for the course, if any.</i>	
L1	Database-System Applications, Purpose of Database Systems, View of Data, Database Languages. Relational Databases	CO1
L2	Database Design, Data Storage and Querying, Transaction Management	CO1
L3	Structure of Relational Databases, Database Schemas, Keys, Schema Diagrams,	CO1
L4/ T1	Database Architecture, Database Users and Administrators	CO1
L5	Relational Query Languages, Relational Operations.	CO1
L6	Relational Algebra –Fundamental Operations	CO1
L7	Additional and Extended Relational Algebra Operations	CO1
L8/ T2	Problems on Relational Algebra	CO1
L9	SQL Data Definition, SQL Data Types and Schemas	CO2
L10	Integrity Constraints	CO2
L11	Structure of SQL Queries, Additional Basic Operations	CO2
L12/T3	Problems on Basic SQL Queries	CO2
L13	Set Operations, Null Values	CO2
L14	Aggregate Functions.	CO2
L15	Nested Subqueries.	CO2
L16/T4	Problems on Complex SQL Queries	CO2
L17	Continuation of Nested Subqueries	CO2
L18	Modification of the Database, Join Expressions.	CO2
L19	Views, Transactions	CO2
L20/T5	Problems on Complex Join Queries and Views	CO2
L21	Overview of the Design Process, The Entity-Relationship Model, Constraints.	CO3
L22	Entity-Relationship Design Issues	CO3
L23	Extended E-R Features	CO3
L24/T6	Removing Redundant Attributes in Entity Sets	CO3
L25	Entity- Relationship Diagrams	CO3
L26	Features of Good Relational Design, Atomic Domains and First Normal Form	CO4
L27	Decomposition Using Functional Dependencies	CO4
L28/T7	Reduction to Relational Schemas	CO3
L29	Functional Dependency Theory	CO4
L30	Finding attribute closure and canonical cover.	CO4
L31	Decomposition and dependency preservation check	CO4
L32/T8	Problems on functional dependency and canonical cover	CO4
L33	Decomposition using Multivalued Dependences	CO4
L34	Algorithms for Decomposition	CO4
L35	File Organization , Organization of Records in Files, B + Tree Index Files	CO1
L36/T9	Tutorial 9-Indexing: Basic Concepts, Ordered Indices	CO1

L37	B+ Tree Extensions, Multiple-Key Access	CO1
L38	Static Hashing, Dynamic Hashing	CO1
L39	Comparison of Ordered Indexing and Hashing, Bitmap Indices	CO1
L40/T10	Transaction Management: Transaction Concept, A simple Transaction model, Transaction Atomicity and Durability	CO5
L41	Transaction Isolation, Serializability	CO5
L42	Transaction Isolation and Atomicity, Transaction Isolation Levels	CO5
L43	Failure Classification, Recovery and Atomicity	CO5
L44/T11	Recovery Algorithm	CO5
L45	Solving Recovery Problems	CO5
L46	Lock-based protocols	CO5
L47	Deadlock handling	CO5
L48/T12	Timestamp based protocols, Validation based protocols	CO5
SDL	QUERY PROCESSING: Overview of Query Processing, Measure of Query Cost, Selection Operation	CO6
SDL	Sorting, Join Operation, Other Operations, NoSQL-Introduction	CO6

Faculty members teaching the course (if multiple sections exist):

Faculty	Section	Faculty	Section
Dr. Dinesh Acharya	CSE A	Ms. Musica Supriya	IT C
Dr. Roopalakshmi R	CSE B	Dr Akshay K C	IT D
Dr. Divya Rao	CSE C	Dr. Girija Attigeri	DS A
Mr Giridhar N Shakarad	CSE D	Ms. Sanjana	DS B
Dr Anup Bhat B	CCE A	Dr. Mayankkumar Lalabhai Patel	DS C
Dr. Chithra K	CCE B	Mr. Anirudhan Adukkathayar C	AIML A
Mr. Vinayak G Pai	CCE C	Dr. Gangothri	AIML B
Dr. Snehal Samanth	CCE D	Ms. Sheena K S	AIML C
Dr. Roshan David Jathanna	IT A	Dr.Soumya Nandan	M &C
Dr. Jashma Suresh	IT B		

References:

Textbooks	1. Silberschatz, Korth, Sudarshan, <i>Database System Concepts</i>, (7e), McGrawHill, New York, 2011. 2. Pramod J Sadalage, Martin Fowler, <i>NoSQL Distilled</i>, Addison-Wesley, 2013.
Self-Directed Learning	

Research Literature/ Case Studies	3. Ramez Elmasri and Shamkant Navathe, Durvasula V L N Somayajulu, Shyam K Gupta, <i>Fundamentals of Database Systems, (6e)</i>, Pearson Education, United States of America, 2011. 4. Thomas Connolly, Carolyn Begg, Database Systems – <i>A Practical Approach to Design, Implementation and Management, (4e)</i>, Pearson Education, England, 2005. 5. Peter Rob, Carlos Coronel, Database Systems–<i>Design, Implementation and Management, (10e)</i>, Course Technology, Boston, 2013
NPTEL/Coursera/any MOOC-based material	Introduction to Database Systems - https://onlinecourses.nptel.ac.in/noc25_cs40/preview

Submitted by:

(Dr.Roopalakshmi.R)
(Signature of the faculty)

Date: 06/01/2026

Approved by:

(Signature of HoD)

Guidelines for Internal Assessment (IA) Components
Academic Term: Jan - May 2026

Guidelines for Internal Assessment (IA) Components

Academic Term: Jan - May 2026

The Internal Assessment (IA) components of theory UG courses for all semesters should align with these guidelines.

Number of components	Minimum 4 (Additional encouraged)
Platform	e-Submission through Brightspace LMS via the Quizzes or Assignments sections.
Pattern	<u>Quizzes Section (With auto-evaluation capabilities)</u> - Multiple Choice / Multi-Select / Arithmetic / Fill in the Blanks / Matching / Ordering / Short Answer / Multi-Short Answer / Written Response (requires manual grading). - Courses involving numerical questions must include Arithmetic questions, where each student is assigned unique (different) value of variables, resulting in distinct questions and answers. - For assessment components that do not include Arithmetic type questions, create a question pool and randomly assign questions to ensure unique sets for each student. - Explore and implement various evaluation criteria like, <i>adding custom weights / correct selections (0.25 pts per answer) / correct answers, limited selections (1pts per correct answer) / right minus wrong selections (± 0.25 pts per answer) / all or nothing / equally weighted</i>, etc. depending on the chosen question format. <u>Assignment Section:</u> Only written responses or e-file uploads are supported with manual evaluation.
Marks and Number of Questions	- The number of questions and marks per question for a component are flexible. - Course coordinators, in consultation with course instructors, may determine these parameters based on the subject's nature and the topics being assessed. - The total marks from all components should be collated and scaled to 20, which must then be entered and released under Final Adjusted Grade in LMS.
Duration	- The duration of each assessment component is flexible and can be decided by course coordinators in collaboration with course instructors, depending on the nature and scope of the questions. - It may vary from a few minutes to several hours or extend to a day or week for assignments / quizzes. - Avoid extremely short-duration MCQ-only quizzes with fixed timing.
Schedule	- The schedule is flexible and can begin from the second week (<i>upon completion of mapping all learners and instructors in the LMS</i>) of the academic calendar. - Ensure assessment components are not scheduled during mid-term week, the last three instructional days, festival holidays, or Institute festival days like Revels. Avoid conducting fixed-time duration assessment components on weekly holidays like Sundays. - Ensure that fixed-time assessments do not overlap with other course quizzes or regular / remedial class timetables. - The schedule in Annexure C may be followed by courses that opt to conduct only four IA components.